

Output Screenshots:

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=====
HYBRID CLOSEST-PAIR OF POINTS EXPERIMENT
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Dataset: OpenFlights Airport Dataset
Number of records: 100
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=====
THRESHOLD EXPERIMENT
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Threshold = 4 Average Time = 0.000189 seconds
Threshold = 8 Average Time = 0.000171 seconds
Threshold = 12 Average Time = 0.000132 seconds
Threshold = 16 Average Time = 0.000135 seconds
Threshold = 20 Average Time = 0.000152 seconds
Threshold = 24 Average Time = 0.000247 seconds
Threshold = 32 Average Time = 0.000173 seconds
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BEST THRESHOLD = 12
Best Average Time = 0.000132 seconds
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FINAL PERFORMANCE COMPARISON
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Input Size: 10

Brute Force : 0.000022 sec | Comparisons: 45
Divide & Conquer: 0.000070 sec | Comparisons: 11
Hybrid (T=12) : 0.000027 sec | Comparisons: 45
Correctness : True

Input Size: 20

Brute Force : 0.000111 sec | Comparisons: 190
Divide & Conquer: 0.000260 sec | Comparisons: 26
Hybrid (T=12) : 0.000044 sec | Comparisons: 91
Correctness : True

Input Size: 40

Brute Force : 0.000386 sec | Comparisons: 780
Divide & Conquer: 0.000238 sec | Comparisons: 43
Hybrid (T=12) : 0.000090 sec | Comparisons: 181
Correctness : True

Input Size: 60

Brute Force : 0.000404 sec | Comparisons: 1770
Divide & Conquer: 0.000386 sec | Comparisons: 69
Hybrid (T=12) : 0.000151 sec | Comparisons: 201
Correctness : True

Input Size: 80

Brute Force : 0.000675 sec | Comparisons: 3160
Divide & Conquer: 0.000495 sec | Comparisons: 101
Hybrid (T=12) : 0.000194 sec | Comparisons: 364
Correctness : True

Input Size: 100

Brute Force : 0.001199 sec | Comparisons: 4950
Divide & Conquer: 0.000793 sec | Comparisons: 142
Hybrid (T=12) : 0.000340 sec | Comparisons: 416

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=====
100 RECORD FINAL OUTPUT
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BRUTE FORCE

Time: 0.001031 seconds

Distance Computations: 4950

Peak Memory: 0.01 KB

Closest Pair:

Ottawa / Gatineau Airport <-> Ottawa Macdonald-Cartier International Airport

Distance: 0.22546189

DIVIDE AND CONQUER

Time: 0.000655 seconds

Distance Computations: 142

Peak Memory: 11.48 KB

Closest Pair:

Ottawa Macdonald-Cartier International Airport <-> Ottawa / Gatineau Airport

Distance: 0.22546189

HYBRID ALGORITHM

Threshold: 12

Time: 0.000275 seconds

Distance Computations: 416

Peak Memory: 10.72 KB

Closest Pair:

Ottawa Macdonald-Cartier International Airport <-> Ottawa / Gatineau Airport

Distance: 0.22546189

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=====
RESULTS SAVED
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Results file: results_IDLE.csv

Experiment completed successfully.

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